

Highlights

PCFormer: A Pose-Context Bottleneck Transformer for School Violence Detection

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- PCFormer fuses pose and context cues for robust school violence detection.
- PCFormer reduces visual noise through compact pose-context representation learning.
- SAVE offers 1,826 real school videos with varied views for domain learning.
- PCFormer surpasses many SOTA methods on SAVE, Movies, Hockey, and Crowd Violence.

PCFormer: A Pose-Context Bottleneck Transformer for School Violence Detection

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Abstract

School violence detection remains challenging due to complex human interactions, diverse surveillance conditions, and the lack of domain-specific datasets. This study introduces PCFormer, a dual-stream pose-context framework for video-based violence detection in educational settings. The pose-guided action stream uses YOLOv11 Pose to extract skeletal keypoints from sampled frames, renders them onto background-free canvases, and encodes the resulting skeleton representations with MobileNet-v3-Small to obtain motion-focused features. In parallel, the semantic context stream employs a Vision Transformer-based action recognition model to extract global contextual cues from RGB frames. To integrate these complementary representations, PCFormer adopts a Multimodal Bottleneck Transformer, where learnable bottleneck tokens mediate compact and controlled interaction between pose-guided action features and contextual features. Gated cross-modal fusion and hierarchical attention pooling are then used to produce a video-level descriptor for binary violence classification. To support research

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in school-specific violence detection, we further introduce SAVE (School Aggression and Violence Evaluation), a dataset of 1,826 video clips recorded in classroom environments from multiple camera viewpoints. Experiments show that PCFormer achieves 93.06% accuracy on SAVE and strong performance on three public benchmarks: Hockey Fight (99.10%), Movies Fight (100.00%), and Crowd Violence (99.60%). The dataset and source code will be made publicly available at:

Keywords: School violence, Vision Transformer, Attention, Violence Detection

1. Introduction

School violence is defined as the intentional use of physical force or power to cause harm—either physical or psychological—within the school environment¹. Such behaviors may arise from interpersonal conflicts among students or result from external actors intruding into the school setting. School violence can manifest in various forms, including both physical and psychological aggression, and is commonly expressed through bullying, intimidation, and coercion, all of which contribute to an atmosphere of fear and insecurity [1]. The impacts of school violence are profound, affecting students’ physical and mental health as well as their overall development. Victims often experience psychological issues such as depression, anxiety, and low self-esteem [2, 3], which can in turn negatively influence their academic performance, social relationships, and long-term quality of life. Alarmingly, school violence is a growing global issue that continues to escalate in both frequency and severity. According to recent statistics, approximately 150 million students aged 13 to 15 have experienced some form of violence in or around school². Each month, one in three students is subjected to bullying; around 36% have engaged in physical fights; and nearly one-third have suffered physical attacks within the past year³. These figures highlight the widespread and persistent nature of the problem. Therefore, preventing school violence is not only an urgent priority but also a sustained, long-term commitment—requiring coordinated interventions, policy support, and community engagement at all

¹<https://www.cdc.gov/youth-violence/about/about-school-violence.html>

²<https://www.unicef.org/protection/violence-against-children>

³<https://www.unesco.org/en/health-education/safe-learning-environments>

levels.

Traditional approaches to preventing school violence often rely on human observation and reporting. However, these methods are frequently ineffective due to fears of retaliation and the lack of continuous monitoring, resulting in many incidents going undetected and leading to serious consequences for the victims [4]. In this context, the rapid advancement of artificial intelligence (AI) has driven significant efforts to develop automated systems for detecting violent behavior [5]. Violence detection in videos has emerged as a highly promising research direction, especially given the exponential growth of video data in recent years [6]. Recent studies have explored features such as RGB images [7], optical flow [8], and skeleton [9] data to improve the effectiveness of violence detection. Deep learning models like CNNs are commonly used to extract spatial features from RGB frames, while RNNs are capable of capturing temporal relationships [10], supporting the recognition of continuous actions. For skeleton data, GCN-based models have also been widely applied [11, 12]; however, they are often sensitive to pose estimation errors and face difficulties when scaling to scenarios involving multiple individuals [13]. RGB-based models are typically vulnerable to background interference and lighting conditions, while most existing architectures remain complex, require substantial computational resources for training, and are challenging to deploy in real-world environments [7]. Furthermore, most current studies have only been conducted on general violence datasets that lack realism or fail to reflect the specific characteristics of school environments, thereby limiting their applicability to the school contexts.

To address these limitations, the PCFormer framework is proposed, an advanced hybrid model that jointly learns from both action-specific and contextual features. The action branch utilizes YOLO11 Pose⁴ to extract skeletal keypoints, generating background-free representations that are encoded via a lightweight MobileNet-v3-Small backbone to capture temporal pose dynamics and suppress irrelevant visual noise. In parallel, the context branch leverages a fine-tuned Vision Transformer (ViT) [14] to extract rich contextual representations, capturing both environmental details and multi-subject interactions within the scene.

To efficiently integrate these heterogeneous modalities, PCFormer introduces a Multimodal Bottleneck Transformer (MBT) that restricts cross-

⁴<https://docs.ultralytics.com/tasks/pose>

stream communication through a small set of learnable bottleneck tokens. This design acts as a structural regularizer, distilling crucial cross-modal dependencies without the quadratic computational cost of full cross-attention. The streams are subsequently unified via Gated Cross-Modal Fusion, which adaptively weights the contributions of action and context features, followed by Hierarchical Attention Pooling (HAP) to aggregate the sequence into a robust video-level descriptor by capturing local temporal salience and global context.

To evaluate this framework and address the scarcity of domain-specific benchmarks, we introduce SAVE (**S**chool **A**ggression and **V**iolence **E**valuation), a newly constructed dataset comprising 1,826 high-quality videos of real-world school settings with diverse camera angles and realistic scenarios. Training on this dataset enables the model to learn domain-specific patterns critical for effective violence detection in educational contexts. Extensive experiments demonstrate that PCFormer not only achieves strong performance on SAVE but also outperforms several state-of-the-art methods on widely used public benchmarks, including Movies [15], Hockey Fights [15], and Crowd Violence [16].

2. Related Work

2.1. Deep Learning-based Violence Detection

The integration of deep learning into video-based violence detection has emerged as a highly promising research direction, attracting widespread attention and yielding notable achievements. Khan et al. [17] introduced VD-Net, a CNN-based surveillance system that achieved remarkable accuracy rates of 98.5% on the Hockey Fight dataset, 99.0% on the Movie dataset, and 97.0% on Violent Flow. Similarly, Haque et al. [7] proposed BrutNet, which combines deep CNNs with GRUs to capture both spatial and temporal dynamics, reaching 100% accuracy on the Movie dataset and over 97% on AVD and Hockey. More recently, Soliman et al. [18] achieved outstanding results by integrating deep CNNs with a voting ensemble strategy, demonstrating improved accuracy and robustness across various datasets, including Hockey Fight, RWF-2000, and Surveillance Video. Despite these promising outcomes, RGB-based methods still face inherent limitations, particularly in scenarios involving complex backgrounds and highly dynamic human motion [19]. Furthermore, such models typically require high computational resources to maintain optimal performance [20].

To address this limitations, recent studies have explored the potential of representing skeletal features through Graph Convolutional Networks (GCNs). The study in [9] proposed three models—GCNOnlyEdge, GCNVertexEdge, and SKE-A3TGCN—in which GCNs are combined with attention mechanisms to learn motion features from graph-based human skeleton representations. Notably, SKE-A3TGCN demonstrates superior performance by effectively capturing dynamic spatial relationships among key joints, significantly improving the accuracy of violent behavior recognition. Experiments on the Hockey Fight, RWF-2000, and Violence in Movies datasets confirm the model’s robustness and high effectiveness in complex surveillance environments. Unlike previous methods that primarily process skeleton sequences over time, Su et al. [19] proposed a novel approach that represents videos as 3D point clouds derived from skeletal pose coordinates. Based on this, the authors introduced the Skeleton Points Interaction Learning (SPIL) module, which models interactions between skeleton points using a spatial-temporal and feature-based weighted distribution strategy. In addition, Janbi et al. [21] developed ESTS-GCNs—an ensemble spatial-temporal skeleton-based graph architecture for violence detection in surveillance videos. The model employs three spatial pipelines (channel-wise topology, self-attention, and graph attention networks) for enhanced motion adaptability. Trained on the Real-Life Violence Situations and NTU-Violence datasets, it achieved nearly 93% accuracy, exceeding previous methods by over 10%.

Despite these advances, GCN-based methods still face several limitations. They are sensitive to minor perturbations in joint coordinates, which can lead to drastically incorrect predictions. The complexity of GCNs also scales with the number of people in a scene, limiting their scalability in multi-person interaction scenarios [13]. Since GAT computes only static attention, it may limit the expressiveness of the attention mechanism in capturing fine-grained, temporally varying interactions [21]. As these models focus solely on body movement patterns, they often overlook contextual information, reducing recognition accuracy in complex environments involving group interactions or dynamic scenes [22].

These limitations open up several compelling directions for future research. First, the representational capacity of skeletal keypoints remains underutilized and warrants deeper exploration, particularly in modeling subtle and coordinated human motions. Second, incorporating contextual cues—such as scene semantics, inter-personal distance, and interaction intent—can play a crucial role in resolving ambiguities in crowded or dynamic scenarios [23].

Finally, multi-stream representation learning that jointly encodes complementary modalities may offer a more holistic and robust understanding of violent behavior across diverse environments [24].

2.2. Violence Dataset

In the field of violence detection, data plays a foundational role in developing and evaluating the effectiveness of deep learning models. Current datasets are constructed from various sources such as surveillance cameras, films, YouTube videos, or self-recorded clips. This diversity allows models to learn a wide range of features related to violent behavior and helps reduce the risk of overfitting. Many large-scale datasets, integrating visual, audio, and motion information, are actively supporting the training of violence detection systems in real-world environments.

Large datasets such as the XD-Violence [28] provide rich resources, particularly for training deep learning models. Additionally, datasets like RWF-2000 [25], Real Life Violence Situations Dataset [27], and Crowd Violence [16] use real-world footage, enabling models to learn the characteristics of violent behavior in practical settings, which increases the applicability of these systems in real-world surveillance. Furthermore, datasets like Movies Fight [15] offer diverse violent scenarios from cinematic content, while XD-Violence [28], Real-Life Violence Situations [27] includes real-life situations ranging from street fights to public disturbances. Most of these datasets maintain a balance between violent and non-violent videos, facilitating more effective model training for classification tasks. Some datasets also include audio, which supports more detailed analysis of violent incidents. Table 1 summarizes the presented related work for this study.

However, these datasets also present several major limitations. Most notably, they are not focused on the school environment, which limits their applicability in developing school surveillance systems. For example, Hockey Fight [15] focuses on sports-related violence, Movies Fight features scripted scenes from films, and Real Life Violence Situations [27] captures street fights. These scenarios do not reflect the nature of school violence, which often involves less severe actions occurring in confined spaces. In addition, some datasets like Movies Fight [15] use staged content that does not accurately represent real-world behavior, especially in educational environments. Violent scenarios in films can differ significantly from reality.

Moreover, datasets such as Movies Fight (200 videos) [15], Crowd Violence (246 videos) [16], and AIRTLab (350 videos) [26] are relatively small

Dataset	Origin	Size	Strengths	Limitations
Hockey Fight [15]	Ice hockey matches	1000 videos (500 violent, 500 non-violent)	Large size, clear examples of physical violence	Sports context, not representative of school scenarios
Movies Fight [15]	Movie fight scenes	200 videos (100 violent, 100 non-violent)	Diverse scenarios	Staged, unrealistic, not suitable for school context
Crowd Violence [16]	YouTube crowd violence videos	246 videos (123 violent, 123 non-violent)	Real-world footage	Small size, unclear school relevance, non-surveillance angles
RWF-2000 [25]	Surveillance cameras	2000 videos (1000 violent, 1000 non-violent)	Large size, real surveillance footage	General surveillance, not school-focused
AIRTLab [26]	Self-recorded clips	350 videos (230 violent, 120 non-violent)	High resolution	Small size, not reflective of school settings
Real-Life Violence Situations [27]	YouTube street fight videos	2000 videos (1000 violent, 1000 non-violent)	Large size, real-world scenes	Focused on street fights, unsuitable for school context
XD-Violence [28]	Various sources including YouTube	4754 videos	Large scale, includes audio	Generic violence, not school-specific
Kranok-NV [29]	Surveillance for normal behavior, online sources for violence	3683 videos	High resolution, static angles for normal actions, practical analysis support	Severely imbalanced, not tailored to school violence
UCF-Crime [30]	Real surveillance videos	1900 videos (950 normal, 950 abnormal)	Large-scale, includes 13 anomaly types (e.g., abuse, robbery)	Long videos, sparse abnormal events, requires long-sequence modeling, not school-specific
Web Abnormality [31]	Web-collected videos	20 sequences (12 normal, 8 abnormal)	Supports anomaly detection in videos	Small size, potential data bias, not focused on school violence
Surveillance Camera Fight [32]	YouTube surveillance fights	300 videos (150 fights, 150 non-fights)	Designed for fight detection, balanced dataset	Small size, short clips (2s), limited scenarios, not school-focused

Table 1: Overview of public violence detection datasets

in size, which is insufficient for training complex deep learning models and limits their generalization capability. Although Crowd Violence [16] uses real-world footage, it remains unclear whether it reflects school-specific situations. Other datasets like RWF-2000 [25] are primarily geared toward general public surveillance rather than educational settings. School violence has unique characteristics, such as occurring in confined spaces or being captured by fixed-position cameras. As a result, existing datasets—despite their diversity in origin and violence scenarios—cannot be directly applied to school violence surveillance systems.

Therefore, the datasets should capture school-related violent behavior using surveillance footage from educational institutions, with appropriate contextual and camera angle considerations. Such datasets will help deep learning models detect violence in schools more accurately and facilitate the practical deployment of security surveillance systems in educational settings.

3. School Aggression and Violence Evaluation Dataset (SAVE)

Based on the analysis presented in Section 2.2, there is currently no publicly available violence dataset specifically developed within the context of school environments. To address this gap and enhance the effectiveness of violence detection models, the SAVE dataset (School Aggression and Violence Evaluation) is constructed to support research and development in this critical area. It is a domain-specific dataset that focuses on classroom scenarios. This study received formal approval from the department chair, authorizing video recording within actual classroom environments. Prior to participation, each student was provided with a comprehensive explanation of the study’s objectives, data collection procedures, and intended use of the recordings, and subsequently gave written informed consent [33]. The consent form explicitly specified the measures taken to protect participant anonymity, the data encryption protocols, the defined data retention period, and the plan for releasing only anonymized video segments ensuring the privacy and rights of all participants throughout the research.

The SAVE dataset is systematically designed and clearly categorized into two primary classes: violent and non-violent behaviors. Each class is further divided into a range of diverse actions to enhance real-world applicability. The violent class encompasses explicit forms of physical aggression, including *choking, kicking, punching, headlocks, knee strikes, hair pulling, slapping*, and *the use of objects to cause harm*. In contrast, the non-violent class in-

cludes typical school-related activities such as *pointing*, *talking*, *walking*, *placing an arm around another person*, and other socially interactive behaviors. To ensure comprehensive coverage and better reflect real-world nuances, the dataset also includes non-violent but physically interactive behaviors—such as *patting on the shoulder*, *high-fives*, and *playful interactions*—allowing the model to more accurately distinguish between harmful and benign physical contact.

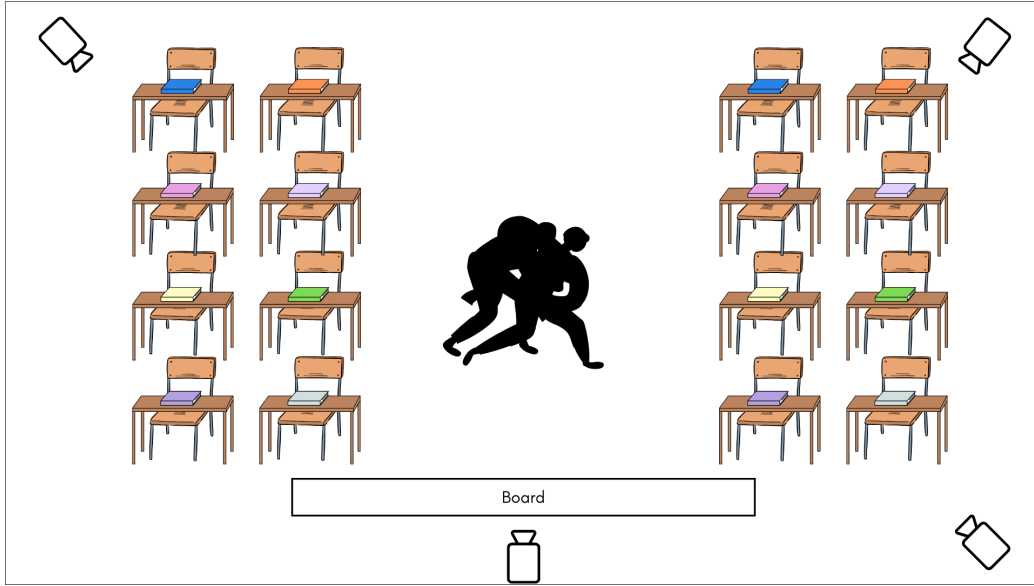


Figure 1: Camera placement setup simulating typical school surveillance viewpoints for multi-angle video capture

To ensure authenticity and accurately reflect the context of school violence, all scenes were recorded in actual classroom environments. To enhance the diversity and richness of the dataset, four cameras were strategically placed to simulate the typical viewpoints of school surveillance systems, as illustrated in Figure 1. The recorded videos vary in resolution (720p and 1080p) and frame rate (30 fps and 60 fps), ensuring both visual clarity and smooth motion representation. To ensure maximum diversity in postures and action execution styles, participants involved in the filming process took turns switching roles in each scene. Subsequently, the recorded videos were carefully segmented into clips ranging from 2 to 5.5 seconds, with each clip capturing the complete progression of an action—from its initiation to full completion.



Figure 2: Representative samples of violent and non-violent actions in the SAVE dataset

Since the participants were not professional actors, a strict evaluation protocol was applied to all clips containing violent behaviors to ensure that the dataset authentically reflects real-world school violence scenarios. This protocol was based on three core criteria: (1) the video must clearly exhibit a specific predefined violent behavior; (2) the behavior must be presented in a clear and unambiguous manner; and (3) the behavior must be executed decisively, forcefully, or in a way that indicates a high degree of potential danger. To ensure objectivity, a peer-review process was conducted by three independent experts. Only the videos that met all three criteria, as agreed upon by all experts, were retained in the final dataset.

After this thorough collection and filtering process, the finalized dataset consists of 1,826 video clips, almost evenly distributed between the two

classes: 921 labeled as violent and 905 labeled as non-violent. In terms of duration, the shortest clip is 2 seconds long, the longest is 5.4 seconds, and the average duration is 3.9 seconds to ensure that each clip captures the full development of the corresponding behavior. Figure 2 showcases representative examples of both violent and non-violent behaviors captured in the dataset. The dataset will be provided upon request.

4. Pose-Context Bottleneck Transformer for School Violence Detection

We propose PCFormer, a context-aware dual-stream framework for video-based school violence detection. Given an input video, PCFormer first samples a compact sequence of frames to reduce temporal redundancy while preserving the progression of human interactions. The sampled frames are then processed by two complementary streams. The pose-guided action stream extracts skeleton-only representations to emphasize human motion and suppress background appearance, whereas the semantic context stream encodes global scene cues from RGB frames. The two feature sequences are projected into a shared latent space and interact through a bottleneck-based transformer, where a small set of learnable tokens constrains cross-modal communication. Finally, gated fusion and hierarchical attention pooling are used to produce a video-level representation for binary violence classification. The overall architecture of the proposed framework is illustrated in Figure 3.

4.1. Temporal Frame Sampling

Video sequences typically contain considerable temporal redundancy, since adjacent frames often exhibit only minor visual variations. Processing every frame may therefore increase computational cost while providing limited additional discriminative information for violence recognition. To reduce this redundancy, we employ a uniform temporal sampling strategy that preserves the progression of human actions while decreasing the number of frames processed by subsequent network components.

Specifically, each video is sampled at a fixed rate of six frames per second. For every one-second temporal interval, the first frame is retained, and the remaining frames are selected at approximately uniform positions within that interval. This design allows the sampled sequence to capture both the onset and temporal evolution of physical interactions, including rapid movements that are often associated with violent behavior. In addition, using a

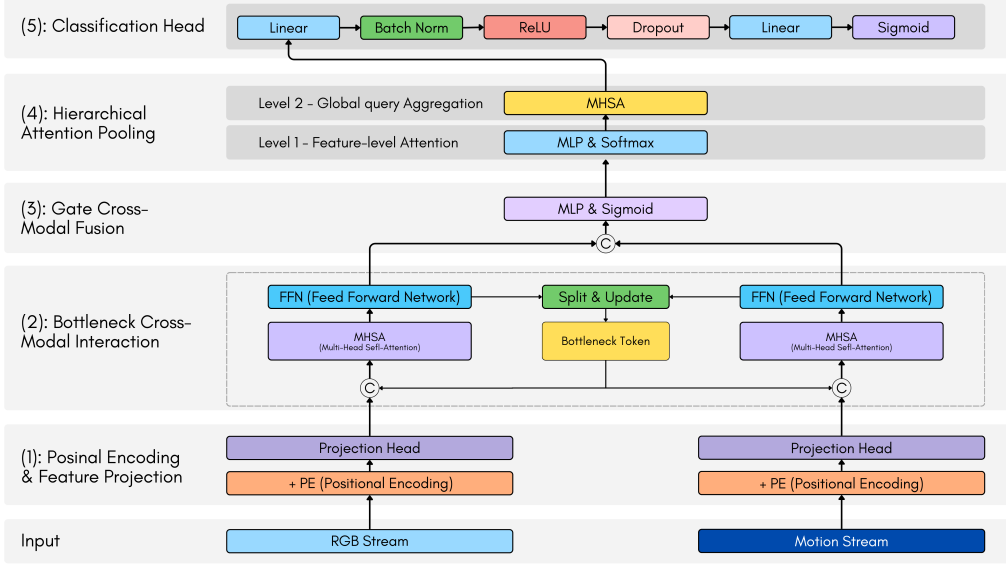


Figure 3: Overall architecture of PCFormer, illustrating the dual-stream design that integrates skeletal-based action features and contextual scene representations for school violence detection.

fixed sampling rate helps normalize videos recorded at different frame rates, thereby providing a more consistent temporal representation across samples.

Formally, let a video be denoted as $V \in \mathbb{R}^{T \times H \times W \times C}$, where T is the total number of frames, and H , W , and C denote the frame height, width, and number of RGB channels, respectively. After temporal sampling, the selected frame sequence is represented as

$$I = \{I_1, I_2, \dots, I_t\}, \quad (1)$$

where $I_i \in \mathbb{R}^{H \times W \times C}$ denotes the i -th sampled frame, $1 \leq i \leq t$, and $t \leq T$. This compact sequence is then used as the input for both the pose-guided action stream and the semantic context stream.

4.2. Dual-Stream Feature Extraction

Violent behavior recognition requires understanding both how individuals move and the contextual circumstances in which those movements occur. Relying solely on appearance-based representations often introduces background distractions, whereas motion-only representations may overlook important situational cues. To address these challenges, PCFormer adopts

a dual-stream feature extraction framework consisting of a pose-guided action stream and a semantic context stream. The action stream focuses on learning motion dynamics from human skeletal structures, while the context stream captures scene-level semantics and interpersonal interactions. Together, these complementary representations provide a more comprehensive understanding of human behavior in complex school environments.

4.2.1. Pose-guided Action Stream

To capture human motion while suppressing irrelevant visual information, the action stream first extracts pose-based representations from each sampled frame. Specifically, the 2D skeletal structures of all visible individuals in each selected frame I_i are estimated using YOLOv11 Pose. Each detected keypoint is represented by its spatial coordinate pair (x, y) and corresponds to a predefined human body joint. From the full pose output, we retain a compact set of 13 motion-relevant joints covering the upper body, torso, and lower body regions.

For a given frame I , let $K_I = \{K_1, K_2, \dots, K_m\}$ denote the set of skeletons detected in the frame, where m is the number of visible individuals and K_p represents the skeleton of person p ($1 \leq p \leq m$). Each skeleton is defined as $K_p = \{(x_j^p, y_j^p) \mid 1 \leq j \leq 13\}$, where (x_j^p, y_j^p) denotes the 2D coordinate of the j -th selected joint of person p . Figure 4 illustrates the simplified 13-keypoint skeleton structure used in the proposed pipeline. YOLOv11 Pose is adopted because it provides an effective balance between pose localization accuracy, inference speed, and multi-person detection capability, making it suitable for surveillance-oriented school violence detection.

After pose estimation, the extracted skeletons are transformed into skeleton-only visual representations. For each frame, a zero-initialized blank canvas with the same spatial resolution as the original frame is created. The selected keypoints and skeleton connections of all detected individuals are then rendered onto this canvas according to the coordinates in K_I . This process produces a transformed frame I' that preserves the spatial layout of human poses while removing background appearance, clothing, texture, and other scene-level distractions.

Applying this transformation to all sampled frames yields a sequence of skeleton-based frames: $I' = \{I'_1, I'_2, \dots, I'_t\}$, where I'_i denotes the skeleton-rendered representation of the sampled frame I_i ($1 \leq i \leq t$). The transformation process is illustrated in Figure 5. Compared with directly using raw keypoint coordinates, rendering skeletons as images allows the model

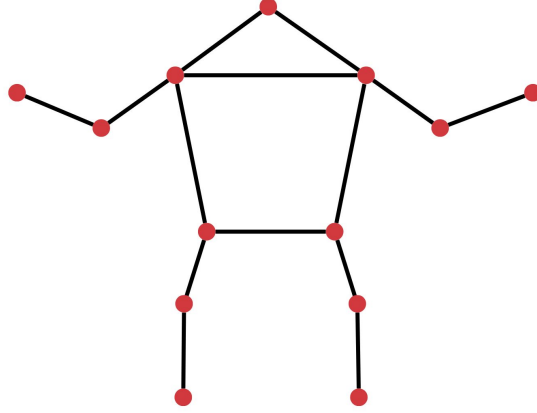


Figure 4: Simplified human skeleton with 13 selected keypoints used for pose-based analysis, as detected by the YOLO11 Pose model

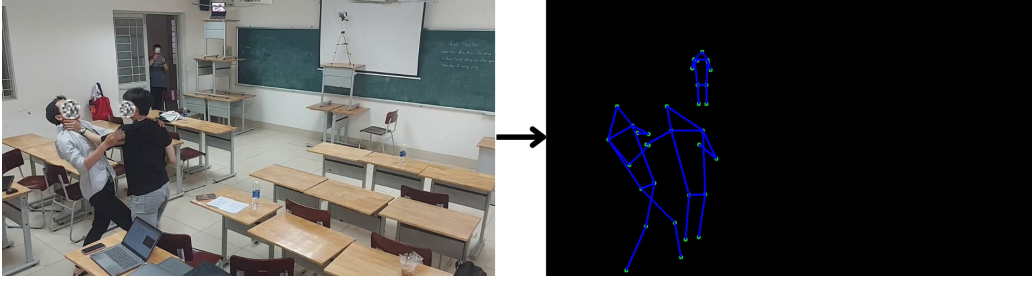


Figure 5: Transformation from raw classroom frame to skeleton-based representation

to exploit spatial patterns using an efficient convolutional backbone while maintaining robustness against irrelevant background variations.

Each skeleton-rendered frame I'_i is then resized to 224×224 pixels and fed into MobileNet-v3-Small [34], which is pre-trained on ImageNet. MobileNet-v3-Small is selected due to its favorable trade-off between representational capacity and computational efficiency, which is important for practical surveillance applications. In this stream, MobileNet-v3-Small extracts frame-level pose representations from the skeleton images, while temporal dependencies among frames are modeled by subsequent sequence-level interaction and aggregation modules.

A global average pooling layer is applied to the output feature map of MobileNet-v3-Small to obtain a compact feature vector for each frame. Let $f_{a_i} \in \mathbb{R}^{576}$ denote the action feature extracted from the transformed frame

I'_i . The resulting action feature sequence is defined as

$$F_a = \{f_{a_1}, f_{a_2}, \dots, f_{a_t}\}, \quad F_a \in \mathbb{R}^{t \times 576} \quad (2)$$

This sequence provides a motion-focused representation of the video and is subsequently combined with contextual features in the cross-modal interaction module.

4.2.2. Semantic Context Stream

Although skeletal representations provide informative cues about human motion, they may be insufficient for distinguishing violent behaviors from visually similar non-violent interactions. Therefore, in addition to the pose-guided action stream, PCFormer incorporates a semantic context stream to extract complementary information from the original RGB frames.

The HARV⁵ model is used as the backbone for contextual feature extraction. HARV is a Vision Transformer (ViT)-based model fine-tuned for human action recognition on the HAR dataset⁶. Since it is optimized for recognizing human actions rather than generic object categories, HARV is expected to provide behavior-relevant visual representations that are suitable for capturing scene-level cues, human presence, and visible interaction patterns in classroom environments.

For each sampled RGB frame I_i , the frame is resized to match the input resolution required by HARV and then forwarded through the ViT encoder. The contextual representation is obtained from the output [CLS] token, which summarizes the global visual information of the frame. Let $f_{c_i} \in \mathbb{R}^{768}$ denote the contextual feature vector extracted from frame I_i . Given the sampled frame sequence $I = \{I_1, I_2, \dots, I_t\}$, the semantic context stream produces a sequence of frame-level contextual embeddings:

$$F_c = \{f_{c_1}, f_{c_2}, \dots, f_{c_t}\}, \quad F_c \in \mathbb{R}^{t \times 768}. \quad (3)$$

These embeddings encode high-level contextual information from the original RGB frames, such as scene layout, surrounding objects, and visual cues related to human interactions. Unlike the pose-guided action stream, which

⁵HARV (Human-Action-Recognition-ViT): <https://huggingface.co/rvv-karma/Human-Action-Recognition-ViT-Base-patch16-224>

⁶HAR (Human Action Recognition): https://huggingface.co/datasets/Bingsu/Human_Action_Recognition

suppresses background appearance to emphasize body motion, the semantic context stream preserves global visual information that may help disambiguate aggressive behaviors from benign physical contact.

The resulting action feature sequence F_a and context feature sequence F_c therefore provide complementary representations of the same video. The former emphasizes pose-based motion patterns, while the latter retains semantic information from the original scene. These two feature sequences are subsequently projected into a shared latent space and jointly modeled through the proposed cross-modal interaction module to learn unified behavior-aware representations.

4.3. Cross-Modal Bottleneck Interaction

The action and context streams provide complementary representations of the same video. The action stream emphasizes pose-based motion patterns, whereas the context stream preserves global semantic cues from the original RGB frames. However, directly applying full cross-attention between all action and context tokens can introduce unnecessary computational cost and may allow unrestricted information exchange, leading to modality dominance or overfitting. To address these issues, PCFormer adopts a Multimodal Bottleneck Transformer (MBT) [35], in which cross-stream communication is mediated by a small set of learnable bottleneck tokens.

Instead of establishing direct pairwise attention between action and context tokens, PCFormer enables the two streams to exchange information through a compact set of learnable bottleneck tokens. This constrained communication mechanism reduces redundant cross-modal interactions, encourages the model to distill task-relevant shared cues, and provides a regularizing effect during representation learning.

Input Projection and Positional Encoding. The action and context features have different dimensionalities, with $\mathbf{F}_a \in \mathbb{R}^{t \times 576}$ and $\mathbf{F}_c \in \mathbb{R}^{t \times 768}$. Therefore, before cross-modal interaction, both feature sequences are projected into a common D -dimensional latent space. Let $d_a = 576$ and $d_c = 768$ denote the original feature dimensions of the action and context streams, respectively. For each stream $k \in \{a, c\}$, the projected representation is computed as

$$\mathbf{H}_k^{(0)} = \text{LayerNorm}(\mathbf{F}_k \mathbf{W}_k + \mathbf{b}_k) \in \mathbb{R}^{t \times D}, \quad (4)$$

where $\mathbf{W}_k \in \mathbb{R}^{d_k \times D}$, $\mathbf{b}_k \in \mathbb{R}^D$, and $D = 256$. To preserve temporal order,

sinusoidal positional encoding is added to each projected sequence:

$$\text{PE}(i, 2j) = \sin\left(\frac{i}{10000^{2j/D}}\right), \quad \text{PE}(i, 2j+1) = \cos\left(\frac{i}{10000^{2j/D}}\right), \quad (5)$$

where i denotes the temporal position and j indexes the feature dimension. The input to the bottleneck interaction module is then obtained as

$$\mathbf{H}_k^{(0)} \leftarrow \text{Dropout}\left(\mathbf{H}_k^{(0)} + \mathbf{PE}\right), \quad k \in \{a, c\}. \quad (6)$$

Multimodal Bottleneck Transformer (MBT) Interaction. To enable controlled interaction between the two streams, we introduce $K = 4$ learnable bottleneck tokens $\mathbf{B}^{(0)} \in \mathbb{R}^{K \times D}$, which are initialized from a normal distribution and expanded along the batch dimension during training and inference. These tokens serve as a shared communication interface between the action and context streams.

At the ℓ -th bottleneck layer, the shared bottleneck tokens are concatenated with the token sequence of each stream:

$$\tilde{\mathbf{H}}_k^{(\ell)} = \left[\mathbf{H}_k^{(\ell)} \parallel \mathbf{B}^{(\ell)}\right] \in \mathbb{R}^{(t+K) \times D}, \quad k \in \{a, c\}. \quad (7)$$

A multi-head self-attention operation with eight heads is then applied independently to each concatenated sequence, followed by a position-wise feed-forward network:

$$\tilde{\mathbf{H}}_k^{(\ell)} \leftarrow \text{LayerNorm}\left(\tilde{\mathbf{H}}_k^{(\ell)} + \text{MHSA}\left(\tilde{\mathbf{H}}_k^{(\ell)}, \tilde{\mathbf{H}}_k^{(\ell)}, \tilde{\mathbf{H}}_k^{(\ell)}\right)\right), \quad (8)$$

$$\tilde{\mathbf{H}}_k^{(\ell)} \leftarrow \text{LayerNorm}\left(\tilde{\mathbf{H}}_k^{(\ell)} + \text{FFN}\left(\tilde{\mathbf{H}}_k^{(\ell)}\right)\right). \quad (9)$$

The feed-forward network is defined as

$$\text{FFN}(\mathbf{x}) = \text{Dropout}\left(\mathbf{W}_2 \text{GELU}\left(\mathbf{W}_1 \mathbf{x}\right)\right), \quad (10)$$

where $\mathbf{W}_1 \in \mathbb{R}^{D \times 256}$ and $\mathbf{W}_2 \in \mathbb{R}^{256 \times D}$.

After the attention and feed-forward operations, the concatenated sequence is split back into two parts. The first t tokens correspond to the updated stream representation, while the remaining K tokens correspond to the stream-specific bottleneck representation:

$$\mathbf{H}_k^{(\ell+1)} = \tilde{\mathbf{H}}_k^{(\ell)}[:, :t, :], \quad \mathbf{B}_k^{(\ell+1)} = \tilde{\mathbf{H}}_k^{(\ell)}[:, t:, :], \quad k \in \{a, c\}. \quad (11)$$

Since the bottleneck tokens are updated separately within each stream, the stream-specific bottleneck states are aggregated to form the shared bottleneck representation for the next layer:

$$\mathbf{B}^{(\ell+1)} = \frac{1}{2} (\mathbf{B}_a^{(\ell+1)} + \mathbf{B}_c^{(\ell+1)}) . \quad (12)$$

This aggregation step allows information from both streams to be exchanged only through the compact set of bottleneck tokens. As a result, the action stream can receive context-aware cues, while the context stream can incorporate motion-aware information, without requiring direct pairwise attention between all tokens from the two modalities.

After L bottleneck layers, the module outputs two cross-modally enhanced feature sequences, $\mathbf{H}_a^{(L)}$ and $\mathbf{H}_c^{(L)}$, which are subsequently passed to the spatiotemporal fusion and aggregation module.

4.4. Cross-Modal Fusion and Temporal Aggregation

Gated Cross-Modal Fusion. After L bottleneck layers, the action and context streams produce two cross-modally enhanced feature sequences, $\mathbf{H}_a^{(L)}, \mathbf{H}_c^{(L)} \in \mathbb{R}^{t \times D}$. Although both streams describe the same video, they capture different types of information: the action stream emphasizes pose-based motion patterns, whereas the context stream preserves semantic cues from the original RGB frames. To adaptively combine these complementary representations, we employ a gated cross-modal fusion mechanism.

For each temporal position i , let $\mathbf{h}_{a_i}, \mathbf{h}_{c_i} \in \mathbb{R}^D$ denote the corresponding action and context tokens. A dimension-wise gate is computed as

$$\mathbf{g}_i = \sigma(\mathbf{W}_g[\mathbf{h}_{a_i} \parallel \mathbf{h}_{c_i}] + \mathbf{b}_g) \in [0, 1]^D, \quad \mathbf{W}_g \in \mathbb{R}^{D \times 2D}, \quad (13)$$

where $\parallel$ denotes feature concatenation and $\sigma(\cdot)$ is the sigmoid function. The fused representation at temporal position i is then obtained by

$$\mathbf{z}_i = \text{LayerNorm}(\mathbf{g}_i \odot \mathbf{h}_{a_i} + (1 - \mathbf{g}_i) \odot \mathbf{h}_{c_i}) \in \mathbb{R}^D, \quad (14)$$

where $\odot$ denotes element-wise multiplication. This gate allows the model to dynamically control the contribution of action and context features at each temporal position and feature dimension. The resulting fused sequence is denoted as

$$\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_t\} \in \mathbb{R}^{t \times D}. \quad (15)$$

Hierarchical Attention Pooling. To aggregate the fused sequence into a compact video-level descriptor, we employ Hierarchical Attention Pooling (HAP). This module consists of two stages: local temporal attention weighting and global query-based aggregation.

Step 1 — Local Temporal Attention Weighting. First, a two-layer attention network computes a scalar importance score for each temporal position:

$$e_i = \mathbf{v}^\top \tanh(\mathbf{W}_a \mathbf{z}_i), \quad \mathbf{W}_a \in \mathbb{R}^{(D/2) \times D}, \quad \mathbf{v} \in \mathbb{R}^{D/2}. \quad (16)$$

The attention weights are then normalized across all temporal positions:

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^t \exp(e_j)}, \quad \sum_{i=1}^t \alpha_i = 1. \quad (17)$$

Each fused token is reweighted according to its temporal importance:

$$\hat{\mathbf{z}}_i = \alpha_i \mathbf{z}_i, \quad (18)$$

and the weighted sequence is represented as

$$\hat{\mathbf{Z}} = \{\hat{\mathbf{z}}_1, \hat{\mathbf{z}}_2, \dots, \hat{\mathbf{z}}_t\} \in \mathbb{R}^{t \times D}. \quad (19)$$

Step 2 — Global Query Aggregation. To obtain a video-level representation, a learnable global query vector $\mathbf{q} \in \mathbb{R}^{1 \times D}$ attends over the weighted sequence $\hat{\mathbf{Z}}$ using multi-head attention with four heads:

$$\mathbf{v}_{\text{pool}} = \text{LayerNorm} \left(\text{MHA} \left(\mathbf{q}, \hat{\mathbf{Z}}, \hat{\mathbf{Z}} \right) \right) \in \mathbb{R}^D. \quad (20)$$

Here, $\mathbf{q}$ serves as a learnable global summarization token, while $\hat{\mathbf{Z}}$ provides the key and value sequences. This hierarchical aggregation allows temporally informative frames to receive higher weights before being summarized into the final video-level descriptor. As a result, frames containing salient physical interactions, abrupt movements, or violence-related cues can contribute more strongly to the final representation used for classification.

4.5. Classification Head and Training Objective

The pooled video-level descriptor $\mathbf{v}_{\text{pool}} \in \mathbb{R}^D$ is passed to a lightweight classification head to predict the probability of violent behavior. The classifier consists of a fully connected layer, Batch Normalization, ReLU activation, Dropout, and a final linear projection followed by a sigmoid function:

$$\mathbf{u} = \text{Dropout} \left(\text{ReLU} \left(\text{BN} \left(\mathbf{W}_1 \mathbf{v}_{\text{pool}} + \mathbf{b}_1 \right) \right) \right), \quad (21)$$

$$\hat{y} = \sigma(\mathbf{W}_2 \mathbf{u} + b_2), \quad (22)$$

where $\mathbf{W}_1 \in \mathbb{R}^{128 \times D}$, $\mathbf{b}_1 \in \mathbb{R}^{128}$, $\mathbf{W}_2 \in \mathbb{R}^{1 \times 128}$, $b_2 \in \mathbb{R}$, and $\hat{y} \in [0, 1]$ denotes the predicted probability that the input video contains violent behavior.

Since the task is formulated as binary classification, the model is optimized using the Binary Cross-Entropy loss:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \quad (23)$$

where N is the number of training samples, $y_i \in \{0, 1\}$ is the ground-truth label, and $\hat{y}_i$ is the predicted probability for the i -th video. During inference, a video is classified as violent if $\hat{y}_i \geq 0.5$ and non-violent otherwise.

5. Experimental results and analysis

5.1. Dataset

To comprehensively evaluate the robustness and adaptability of the proposed model across diverse scenarios, it was independently trained and tested on several established benchmark datasets in addition to SAVE. These datasets include Hockey Fight, Movies Fight, and Crowd Violence, all of which are widely adopted in the field of violence detection research.

The SAVE dataset was divided into 60% training, 20% validation, and 20% testing sets. For the Hockey Fight, Movies Fight, and Crowd Violence datasets, an 80%/20% train-test split was employed [10, 20]. Representative frames from the Hockey Fight, Movies Fight, and Crowd Violence datasets are shown in Figure 6.

5.2. Experimental Setup and Evaluation Metrics

The model is trained on a system with a CPU (29GB RAM) and two T4 GPUs (15GB RAM each). The experiments were conducted with a batch size of 16, and the model was trained for 50 epochs with a learning rate of 1e-4 and a weight decay of 1e-4, using the AdamW optimization algorithm and the Cross Entropy loss function.

The results of each model on the test set are presented in a 2×2 confusion matrix. Based on each confusion matrix, the corresponding evaluation metrics are computed, including Accuracy, Recall, Precision, and F1-Score.



Figure 6: Some sample frames from the Movies (a), Crowd Violence (b), and Hockey Fight (c) datasets.

These values are calculated using equations (24), (25), (26), and (27), where TP stands for true positives, TN for true negatives, FP for false positives, and FN for false negatives.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (24)$$

$$Precision = \frac{TP}{TP + FP} \quad (25)$$

$$Recall = \frac{TP}{TP + FN} \quad (26)$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (27)$$

5.3. Comparison with the State of the Art

5.3.1. SAVE Dataset

As shown in Table 2, PCFormer stands out with exceptional performance, achieving an Accuracy of 93.06%, Recall of 93.08%, Precision of 93.21%, and an F1-Score of 93.13%. This demonstrates that PCFormer not only delivers the highest classification accuracy but also excels in detecting violent

Table 2: Performance comparison of models on the SAVE dataset

Study	Accuracy	Recall	Precision	F1-Score
ViViT [36]	66.67	35.68	95.65	51.97
3D CNN [18]	73.22	80.54	70.62	75.25
BrutNet [7]	79.78	69.73	87.76	77.71
SepConvLSTM [10]	91.80	90.27	93.30	91.76
PCFormer (Ours)	93.06	93.08	93.21	93.13

situations while maintaining a good balance between Recall and Precision, making it an optimal choice for applications requiring high reliability.

Conversely, traditional convolutional approaches like 3D CNN [18] and BrutNet [7] demonstrate moderate performance. While the 3D CNN architecture achieves a reasonable Recall of 80.54%, its low Precision (70.62%) indicates a high propensity for false positives. This suggests that standard 3D convolutions struggle to differentiate between high-energy non-violent activities and actual violence. BrutNet exhibits an opposite trend; it achieves better Precision (87.76%) and overall Accuracy (79.78%) but suffers a considerable drop in Recall (69.73%), rendering it overly conservative and prone to missing genuine violent occurrences. Both models highlight the limitations of standard spatiotemporal convolutional operators in capturing the complex, long-range temporal dependencies inherent in prolonged school violence scenarios.

Among the baseline methods, SepConvLSTM [10] emerges as the most competitive alternative, yielding strong results across all metrics, including an Accuracy of 91.80% and an F1-Score of 91.76%. By effectively coupling spatial convolutions with recurrent temporal modeling, SepConvLSTM captures the sequential dynamics of violent actions far better than pure 3D CNN variants. Nevertheless, it falls short of our proposed PCFormer framework by notable margins of 1.26% in Accuracy and 2.81% in Recall. Although SepConvLSTM marginally edges out PCFormer in Precision (93.30% vs. 93.21%), PCFormer’s significantly higher Recall (93.08% vs. 90.27%) ensures that substantially fewer violent incidents are overlooked—a critical requirement for real-world automated surveillance systems.

Although ViViT [36] is a powerful transformer-based architecture that achieves remarkable results on large-scale and diverse benchmarks such as

Kinetics and Something-Something V2, its performance on the SAVE dataset drops substantially, reaching only 66.67% in Accuracy and 51.97 in F1-Score. In the context of school violence detection, the distinguishing features between violent and non-violent actions are frequently nuanced—characterized by micro-movements, interpersonal spatial relations, and behavioral intent rather than overt physical actions. ViViT, despite its strong spatiotemporal representation capability, tends to overfit to visually dominant cues and lacks the fine-grained behavioral understanding necessary for such sensitive domains. This explains its high Precision (95.65%) but very low Recall (35.68%), suggesting that it identifies only the most obvious violent cases while missing many real instances of aggression.

The superior overall performance of PCFormer, evidenced by its peak F1-Score of 93.13%, confirms that its underlying architecture addresses the core weaknesses of previous methods. Unlike the rigid sequential processing of LSTMs or the unconstrained attention of ViViT, PCFormer’s multimodal fusion and hierarchical aggregation strategies act as robust filters, effectively suppressing background noise while dynamically emphasizing salient violent cues.

5.3.2. Movies Dataset

Table 3: Performance comparison of studies and models on the Movies dataset

Study	Accuracy	Recall	Precision	F1-Score
Su et al., 2020 [19]	98.50	-	-	-
Islam et al., 2021 [10]	100.00	-	-	-
Hachiuma et al., 2023 [12]	99.00	-	-	-
Khan et al., 2024 [17]	99.00	-	-	98.00
Dündar et al., 2024 [18]	100.00	-	-	-
Tran et al., 2025 [9]	100.00	100.00	100.00	100.00
PCFormer (Ours)	100.00	100.00	100.00	100.00

As illustrated in Table 3, the Movies dataset presents a highly saturated benchmark where identifying violence is relatively straightforward due to the overt, staged nature of the actions and clean camera angles. Modern architectures, including our proposed PCFormer framework, as well as recent works by Dündar et al. [18] and Tran et al. [9], consistently reach the

performance ceiling, achieving perfect scores (100.00%) across all metrics. While this dataset no longer poses a significant challenge for contemporary deep learning models, achieving the absolute maximum score serves as a crucial sanity check, verifying PCFormer’s robust foundational capability in accurately modeling fundamental violent spatiotemporal patterns without introducing false negatives or false positives.

5.3.3. Hockey Fight Dataset

Table 4 details the comparative evaluation on the Hockey Fight dataset, characterized by fast-paced motions and highly uniform backgrounds (the ice rink). PCFormer demonstrates highly competitive performance, achieving an Accuracy of 99.10% and establishing state-of-the-art results among fully reported metrics with a Recall of 99.60%, a Precision of 98.62%, and an F1-Score of 99.10%. While prior studies such as Islam et al. [10] and Hachiuma et al. [12] report a marginally higher Accuracy (99.50%), these works omit comprehensive evaluation metrics (Precision, Recall, F1), limiting the assessment of their practical reliability. Among recent comprehensive studies, PCFormer significantly outperforms the model proposed by Tran et al. [9] (99.10% vs. 92.02% in F1-Score) and improves upon the latest work by Negre et al. [37]. The exceptionally high Recall (99.60%) underscores the effectiveness of PCFormer’s hierarchical attention mechanism in continuously tracking rapid, aggressive interactions without losing the temporal thread of the altercation.

Table 4: Performance comparison of studies and models on the Hockey Fight dataset

Study	Accuracy	Recall	Precision	F1-Score
Su et al., 2020 [19]	96.80	-	-	-
Islam et al., 2021 [10]	99.50	-	-	-
Hachiuma et al., 2023 [12]	99.50	-	-	-
Dündar et al., 2024 [18]	95.50	-	-	-
Khan et al., 2024 [17]	98.50	-	-	98.00
Tran et al., 2025 [9]	90.43	91.54	92.61	92.02
Khan et al., 2025 [20]	97.20	95.10	95.10	95.10
Negre et al., 2026 [37]	97.00	98.00	95.00	97.00
PCFormer (Ours)	99.10	99.60	98.62	99.10

5.3.4. Crowd Violence Dataset

The Crowd Violence dataset poses one of the most rigorous challenges in the domain due to severe visual occlusions, complex dynamic backgrounds, and the chaotic nature of multi-person altercations. As demonstrated in Table 5, PCFormer establishes a definitive new state-of-the-art on this benchmark, overwhelmingly outperforming all existing baseline methods. Specifically, PCFormer achieves a near-perfect Accuracy of 99.60% and an F1-Score of 99.59%, substantially surpassing recent models like Khan et al. [17] (97.00% Accuracy) and Negre et al. [37] (90.00% Accuracy).

Most notably, PCFormer yields a flawless Precision of 100.00% paired with an outstanding Recall of 99.20%. This indicates an unprecedented ability to completely eliminate false positives—a common pitfall for traditional models when analyzing dense crowds where sudden non-violent crowd movements (e.g., cheering, running) are easily misclassified as violence. This breakthrough performance can be directly attributed to our proposed Gated Cross-Modal Fusion and bottleneck tokens, which function as an adaptive filter to suppress irrelevant background crowd noise while dynamically isolating the highly specific multimodal cues of genuine physical aggression.

Table 5: Performance comparison of studies and models on the Crowd Violence dataset

Study	Accuracy	Recall	Precision	F1-Score
Varghese et al., 2020 [38]	92.90	92.00	93.00	-
Su et al., 2020 [19]	94.50	-	-	-
Hachiuma et al., 2023 [12]	94.70	-	-	-
Khan et al., 2024 [17]	97.00	-	-	96.00
Negre et al., 2026 [37]	90.00	97.00	86.00	91.00
PCFormer	99.60	99.20	100.00	99.59

5.4. Ablation Study

To rigorously validate the architectural decisions in the PCFormer framework, we conducted extensive ablation studies on the SAVE dataset. Each component of the pipeline was isolated and evaluated, and the results represent the mean and standard deviation across five independent runs to ensure statistical reliability.

Input Stream Modality. Table 6 demonstrates the necessity of the dual-stream approach. The single-stream motion model (MobileNet-v3-Small on skeletons) achieves an accuracy of 91.26%, significantly outperforming the single-stream RGB context model (89.13%). This confirms that human pose dynamics are the primary indicators of violent behavior. However, the motion stream alone lacks scene awareness and struggles with ambiguous interactions (e.g., a friendly hug vs. a wrestling hold). By integrating both streams via our proposed fusion mechanism, the dual-stream model achieves a substantial performance leap to 93.06% accuracy. The context stream provides crucial environmental cues that resolve the ambiguities in the skeleton stream, proving that action and context are highly complementary.

Table 6: Quantitative comparison of the proposed dual-stream approach against single-stream baselines.

Stream mode	Acc.	Pre.	Re.	F1
RGB	0.8913 ± 0.0053	0.8922 ± 0.0253	0.8941 ± 0.0218	0.8926 ± 0.0026
Motion	0.9126 ± 0.0100	0.9279 ± 0.0185	0.8973 ± 0.0251	0.9120 ± 0.0108
Dual (proposed)	0.9306 ± 0.0037	0.9321 ± 0.0114	0.9308 ± 0.0150	0.9313 ± 0.0040

Cross-Modal Interaction Mechanism. A critical innovation of PCFormer is the Multimodal Bottleneck Transformer (MBT). Table 7 compares MBT against a no-interaction baseline (late fusion only) and a standard vanilla cross-attention mechanism. Surprisingly, vanilla cross-attention (91.42%) yields no improvement over the no-interaction baseline (91.42%). In small to medium-scale datasets like SAVE, full bidirectional cross-attention often struggles to align heterogeneous modalities and is highly prone to overfitting, as every token can attend to every other token. In contrast, MBT (93.06%) acts as a powerful structural regularizer. By forcing all cross-stream communication to route through a tightly constrained set of $K = 4$ shared tokens, MBT prevents modality dominance and encourages the network to distill only the most salient, generalizable cross-modal dependencies.

Table 7: Ablation study evaluating the effectiveness of the proposed MBT architecture. Performance is compared against vanilla cross-attention and a baseline without cross-attention.

Interaction mode	Acc.	Pre.	Re.	F1
Vanilla cross attention	0.9142 ± 0.0168	0.9236 ± 0.0365	0.9070 ± 0.0269	0.9146 ± 0.0156
W/o cross attention	0.9142 ± 0.0153	0.9199 ± 0.0246	0.9103 ± 0.0228	0.9147 ± 0.0150
MBT (proposed)	0.9306 ± 0.0037	0.9321 ± 0.0114	0.9308 ± 0.0150	0.9313 ± 0.0040

A key advantage of the proposed MBT architecture is its ability to achieve superior classification performance while maintaining a significantly lower parameter count compared to the vanilla cross-attention mechanism. Specifically, the MBT design yields a 30.6% reduction in the overall model parameters (2.39M vs. 3.45M). Despite this substantial reduction in model complexity, MBT consistently outperforms the vanilla cross-attention baseline, demonstrating a notable improvement of +1.64% in accuracy on the SAVE dataset (93.06% vs. 91.42%). This finding strongly suggests that the bottleneck mechanism effectively acts as an implicit regularizer—filtering out redundant cross-modal noise—rather than a restrictive capacity constraint.

Feature Fusion Strategy. Following the cross-modal interaction layers, the distinct feature streams must be aggregated into a unified representation. As detailed in Table 8, standard deterministic fusion operations—namely addition, multiplication, and concatenation—yield sub-optimal performance, with classification accuracies plateauing between 91.3% and 91.5%. The primary limitation of these conventional methods lies in their assumption of a static inter-modal relationship, effectively treating all temporal frames with equal importance. In contrast, our proposed Gated Cross-Modal Fusion strategy overcomes this limitation, achieving a superior accuracy of 93.06%. By learning to dynamically calibrate the contributions of the action and context streams on a per-dimension basis, the gating mechanism functions as an adaptive filter. Consequently, it selectively amplifies skeletal motion cues during pronounced physical exchanges, while prioritizing broader contextual features when the core action is visually obscured or ambiguous.

Table 8: Ablation study on the effect of different fusion modes. The proposed gated fusion strategy is compared against standard operations including addition, multiplication, and concatenation.

Fusion mode	Acc.	Pre.	Re.	F1
Add	0.9153 ± 0.0171	0.9295 ± 0.0307	0.9016 ± 0.0164	0.9151 ± 0.0166
Multiply	0.9137 ± 0.0105	0.9290 ± 0.0207	0.8984 ± 0.0181	0.9132 ± 0.0104
Concat	0.9137 ± 0.0235	0.9234 ± 0.0473	0.9081 ± 0.0520	0.9139 ± 0.0236
Gated (proposed)	0.9306 ± 0.0037	0.9321 ± 0.0114	0.9308 ± 0.0150	0.9313 ± 0.0040

Temporal Aggregation. School violence is inherently a temporally sparse phenomenon, often occupying only a marginal fraction of a given video sequence. As demonstrated in Table 9, the choice of temporal pooling strategy profoundly impacts model efficacy. Conventional Global Average Pooling (GAP) yields sub-optimal results (89.78% accuracy) as it inadver-

tently dilutes the salient features of transient violent actions with those of the dominant, non-violent background frames. Conversely, max pooling exhibits improved performance (91.64%) by successfully capturing peak activation responses; however, it inherently discards the critical temporal evolution of the scene. Our proposed Hierarchical Attention Pooling explicitly mitigates these limitations, achieving a peak accuracy of 93.06%. By leveraging a local frame-level attention mechanism, the network assigns elevated weights to temporal segments exhibiting sudden motion or aggression. Subsequently, a global query-based multi-head attention module selectively aggregates these highly activated segments, culminating in a robust and discriminative video-level descriptor.

Table 9: Ablation study on the effect of different pooling modes. The proposed hierarchical pooling strategy is compared against standard global average pooling (GAP) and max pooling operations.

Pooling mode	Acc.	Pre.	Re.	F1
GAP	0.8978 ± 0.0063	0.8969 ± 0.0215	0.9027 ± 0.0320	0.8992 ± 0.0077
Max	0.9164 ± 0.0252	0.9138 ± 0.0308	0.9222 ± 0.0258	0.9178 ± 0.0244
Hierarchical (proposed)	0.9306 ± 0.0037	0.9321 ± 0.0114	0.9308 ± 0.0150	0.9313 ± 0.0040

Table 10: Ablation study on the impact of the number of bottleneck tokens. The proposed configuration with 4 bottlenecks achieves the best overall performance.

Token	Acc.	Pre.	Re.	F1
2	0.9230 ± 0.0099	0.9255 ± 0.0306	0.9232 ± 0.0204	0.9238 ± 0.0085
4	0.9306 ± 0.0037	0.9321 ± 0.0114	0.9308 ± 0.0150	0.9313 ± 0.0040
6	0.9120 ± 0.0053	0.9517 ± 0.0083	0.8703 ± 0.0183	0.9090 ± 0.0066
8	0.9000 ± 0.0079	0.9449 ± 0.0263	0.8530 ± 0.0266	0.8960 ± 0.0084

Bottleneck Token Count. The hyperparameter K , representing the number of bottleneck tokens, inherently governs the informational bandwidth of cross-modal communication. As delineated in Table 10, setting $K = 4$ yields the optimal configuration, achieving a peak accuracy of 93.06%. When this bandwidth is excessively constricted ($K = 2$), the limited token capacity restricts the transmission of sufficient contextual cues. Conversely, an overly expansive bottleneck ($K \in \{6, 8\}$) severely diminishes the architectural regularizing effect. Notably, while configurations with $K = 6$ and $K = 8$ report disproportionately high precision (exceeding 94%), they suffer a drastic degradation in recall (87.03% and 85.30%, respectively). This disparity suggests that an excessively wide bottleneck induces the model to overfit

to highly specific multimodal co-occurrences, rendering its decision boundary overly conservative and ultimately causing it to misclassify genuine violent instances as false negatives. Consequently, $K = 4$ establishes the optimal trade-off between robust information flow and effective regularization.

6. Discussion

The experimental results consistently validate our core hypothesis: effective violence detection in complex, dense environments like schools requires the synergistic analysis of both precise skeletal dynamics and broader contextual scene awareness. As demonstrated in the ablation studies, the introduction of the Multimodal Bottleneck Transformer (MBT) is crucial not only for feature fusion but also as a structural regularizer. By bottlenecking the cross-modal communication, the network is forced to distill only the most salient behavioral cues, effectively preventing the model from overfitting to irrelevant visual noise such as crowd movements or static classroom objects.

Furthermore, the introduction and subsequent performance on the SAVE dataset underscore a critical gap in the existing literature. While current deep learning models easily saturate general-purpose datasets like Movies Fight (achieving 100% accuracy), they struggle with the nuanced, confined interactions typical of educational settings. SAVE provides a necessary, challenging benchmark that forces models to distinguish between benign physical interactions (e.g., horseplay, hugs) and actual aggression.

Despite these promising advancements, this study presents certain limitations. First, while the SAVE dataset is extensive, the scenarios were recorded under controlled conditions and may not fully encompass the extreme variations in lighting, motion blur, and camera degradation found in legacy school CCTV systems. Second, the current framework is purely vision-based. In many real-world altercations, audio cues (e.g., shouting, aggressive language) precede physical violence; omitting this modality limits the system’s predictive capability. Finally, the integration of a Vision Transformer in the context stream, coupled with the dual-branch architecture, introduces significant computational overhead. This complexity currently restricts the model’s deployment on low-power edge devices (such as edge-AI cameras), necessitating high-end hardware for real-time inference.

7. Conclusion and Future Work

This study presents PCFormer, a novel dual-stream deep learning framework for school violence detection. The proposed architecture integrates YOLOv11-based skeletal motion representations with Vision Transformer derived contextual features through a Multimodal Bottleneck Transformer, enabling effective discrimination of violent interactions from complex and dynamic classroom environments. To support this task, we introduce the SAVE dataset, a dedicated benchmark specifically designed for school violence detection in educational settings. Extensive experiments conducted on four public benchmark datasets demonstrate the effectiveness and generalizability of PCFormer, achieving state-of-the-art performance with accuracies of 93.06% on SAVE and 99.60% on the challenging Crowd Violence dataset.

Future work will focus on improving the practicality and scalability of the proposed framework. In particular, we will investigate model compression and optimization strategies, including network pruning, quantization, and knowledge distillation, to facilitate deployment on resource-constrained edge surveillance devices. Furthermore, we plan to incorporate complementary modalities, such as audio signals and Vision-Language Models, to enhance contextual reasoning and capture richer semantic information associated with violent incidents. Finally, the SAVE dataset will be expanded to include a broader range of educational environments, such as playgrounds, hallways, and cafeterias, thereby providing a more comprehensive benchmark and supporting the development of robust real-world school safety monitoring systems.

Acknowledgment

This research is funded by Vietnam National Foundation for Science and Technology Development (NAFOSTED) under grant number 102.01-2025.61.

References

- [1] M. Furlong, G. Morrison, The school in school violence: Definitions and facts, *Journal of Emotional and Behavioral Disorders* 8 (2000) 71–82.
- [2] N. Tran, P. Ta, H. Nguyen, H. D. Nguyen, A.-C. Le, Hybrid contextual and sentiment-based machine learning model for identifying depression risk in social media, *Expert Systems with Applications* 291 (2025) 128505.

- [3] P. Ta, N. Tran, H. Nguyen, H. D. Nguyen, Detecting signs of depression on social media: A machine learning analysis and evaluation, *Sustainable Futures* 10 (2025) 100827. doi:10.1016/j.sftr.2025.100827.
- [4] Y. B. Yablon, Students' reports of severe violence in school as a tool for early detection and prevention, *Child development* 88 (2017) 55–67.
- [5] Y. Tang, Y. Chen, S. A. Sharifuzzaman, T. Li, An automatic fine-grained violence detection system for animation based on modified faster r-cnn, *Expert systems with applications* 237 (2024) 121691.
- [6] P. Wu, X. Liu, J. Liu, Weakly supervised audio-visual violence detection, *IEEE Transactions on Multimedia* 25 (2022) 1674–1685.
- [7] M. Haque, H. Nyeem, S. Afsha, Brutnet: A novel approach for violence detection and classification using dcnn with gru, *The Journal of Engineering* 2024 (2024) e12375.
- [8] H. Mohammadi, E. Nazerfard, Video violence recognition and localization using a semi-supervised hard attention model, *Expert Systems with Applications* 212 (2023) 118791.
- [9] N. Tran, H. Nguyen, D. Ly, K. Ngo, H. D. Nguyen, Advancing violence detection with graph-based skeleton motion analysis, *SN Computer Science* 6 (2025) 595.
- [10] Z. Islam, M. Rukonuzzaman, R. Ahmed, M. H. Kabir, M. Farazi, Efficient two-stream network for violence detection using separable convolutional lstm, in: *2021 International Joint Conference on Neural Networks (IJCNN)*, IEEE, 2021, pp. 1–8.
- [11] N. Tran, H. Nguyen, D. Ly, H. D. Nguyen, Violence detection using skeleton data with graph convolutional networks, in: *International Conference on Intelligent Systems and Data Science*, Springer, 2024, pp. 86–97.
- [12] R. Hachiuma, F. Sato, T. Sekii, Unified keypoint-based action recognition framework via structured keypoint pooling, in: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 22962–22971.

- [13] H. Duan, Y. Zhao, K. Chen, D. Lin, B. Dai, Revisiting skeleton-based action recognition, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2022, pp. 2969–2978.
- [14] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby, An image is worth 16x16 words: Transformers for image recognition at scale, in: International Conference on Learning Representations, 2021.
- [15] E. Bermejo Nievas, O. Deniz Suarez, G. Bueno García, R. Sukthankar, Violence detection in video using computer vision techniques, in: Computer Analysis of Images and Patterns: 14th International Conference, CAIP 2011, Seville, Spain, August 29-31, 2011, Proceedings, Part II 14, Springer, 2011, pp. 332–339.
- [16] T. Hassner, Y. Itcher, O. Kliper-Gross, Violent flows: Real-time detection of violent crowd behavior, in: 2012 IEEE computer society conference on computer vision and pattern recognition workshops, IEEE, 2012, pp. 1–6.
- [17] M. Khan, A. El Saddik, W. Gueaieb, G. De Masi, F. Karray, Vd-net: An edge vision-based surveillance system for violence detection, IEEE Access 12 (2024) 43796–43808.
- [18] N. Dünder, A. S. Keçeli, A. Kaya, H. Sever, A shallow 3d convolutional neural network for violence detection in videos, Egyptian Informatics Journal 26 (2024) 100455.
- [19] Y. Su, G. Lin, J. Zhu, Q. Wu, Human interaction learning on 3d skeleton point clouds for video violence recognition, in: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part IV 16, Springer, 2020, pp. 74–90.
- [20] H. Khan, X. Yuan, L. Qingge, K. Roy, Violence detection from industrial surveillance videos using deep learning, IEEE Access (2025).
- [21] N. F. Janbi, M. A. Ghaseb, A. A. Almazroi, Ests-gcn: An ensemble spatial–temporal skeleton-based graph convolutional networks for violence detection, International Journal of Intelligent Systems 2024 (2024) 2323337.

- [22] T. Li, Z. Sun, X. Chen, Group-skeleton-based human action recognition in complex events, in: Proceedings of the 28th ACM International Conference on Multimedia, 2020, pp. 4703–4707.
- [23] N. Tran, T. Le, P. Ta, H. Nguyen, H. D. Nguyen, Contrastive context-aware emotion recognition: A multimodal framework for learning contextualized emotional representations, *Neurocomputing* (2026) 133237.
- [24] H. Nguyen, N. Tran, M. Nguyen, P. Ta, H. D. Nguyen, Attention mechanisms for context-aware emotion recognition, *CCF Transactions on Pervasive Computing and Interaction* (2026) 1–17.
- [25] M. Cheng, K. Cai, M. Li, Rwf-2000: An open large scale video database for violence detection, in: 2020 25th International Conference on Pattern Recognition (ICPR), IEEE, 2021, pp. 4183–4190.
- [26] P. Sernani, N. Falcionelli, S. Tomassini, P. Contardo, A. F. Dragoni, Deep learning for automatic violence detection: Tests on the airtlab dataset, *IEEE Access* 9 (2021) 160580–160595.
- [27] M. M. Soliman, M. H. Kamal, M. A. E.-M. Nashed, Y. M. Mostafa, B. S. Chawky, D. Khattab, Violence recognition from videos using deep learning techniques, in: 2019 ninth international conference on intelligent computing and information systems (ICICIS), IEEE, 2019, pp. 80–85.
- [28] P. Wu, J. Liu, Y. Shi, Y. Sun, F. Shao, Z. Wu, Z. Yang, Not only look, but also listen: Learning multimodal violence detection under weak supervision, in: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXX 16, Springer, 2020, pp. 322–339.
- [29] K. B. Kwan-Loo, J. C. Ortíz-Bayliss, S. E. Conant-Pablos, H. Terashima-Marín, P. Rad, Detection of violent behavior using neural networks and pose estimation, *IEEE Access* 10 (2022) 86339–86352.
- [30] W. Sultani, C. Chen, M. Shah, Real-world anomaly detection in surveillance videos, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2018, pp. 6479–6488.

- [31] R. Mehran, A. Oyama, M. Shah, Abnormal crowd behavior detection using social force model, in: 2009 IEEE conference on computer vision and pattern recognition, IEEE, 2009, pp. 935–942.
- [32] Ş. Aktı, G. A. Tataroğlu, H. K. Ekenel, Vision-based fight detection from surveillance cameras, in: 2019 Ninth International Conference on Image Processing Theory, Tools and Applications (IPTA), IEEE, 2019, pp. 1–6.
- [33] N. Tran, D. Ly, P. Ta, H. Nguyen, H. D. Nguyen, Ordinal-aware multi-modal engagement recognition for collaborative learning, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2026, pp. 2424–2433.
- [34] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, et al., Searching for mobilenetv3, in: Proceedings of the IEEE/CVF international conference on computer vision, 2019, pp. 1314–1324.
- [35] A. Nagrani, S. Yang, A. Arnab, A. Jansen, C. Schmid, C. Sun, Attention bottlenecks for multimodal fusion, *Advances in neural information processing systems* 34 (2021) 14200–14213.
- [36] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lučić, C. Schmid, Vivit: A video vision transformer, in: Proceedings of the IEEE/CVF international conference on computer vision, 2021, pp. 6836–6846.
- [37] P. Negre, R. S. Alonso, J. Prieto, O. García, Video violence detection using pre-trained vgg19 combined with manual logic, lstm layers and bi-lstm layers, *Applied Intelligence* 56 (2026) 72.
- [38] E. B. Varghese, S. M. Thampi, S. Berretti, A psychologically inspired fuzzy cognitive deep learning framework to predict crowd behavior, *IEEE Transactions on Affective Computing* 13 (2020) 1005–1022.